

Real-Time Expiry Alert System for Safer Retail using Barcode and QR Code Recognition.

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Abstract— A real-time expiry detection system integrates advanced technologies such as image processing, neural networks, and database management to deliver accurate and time-efficient expiration analysis. Instead of using preloaded databases, the system leverages barcode and Quick Response code scanning to retrieve product details such as expiry dates. The proposed system has accuracy in retrieving expiry details with packaging by combining Optical Character Recognition as well as image processing methods and classifying products of their shelf life. When identifying an expired or near-expiry product, an automatic alert is created at the point of sale that notifies the retailers and clients in real time. The technology also includes a user- friendly dashboard that enables retailers to easily track and manage the expiry status of their products, reducing the chance for human error and maintaining compliance with safety standards. This approach leads to centering retail through automating expiry detection in stores, it reduces food spoilage and loss, and saves consumer health making the store a well-organized, safe and a more hygienic place to shop.

Keywords— Barcode Scanning, Expiry Detection, Optical Character Recognition, Quick Response Code Processing, Real-Time Monitoring, Retail Safety, Smart Inventory Management.

I. INTRODUCTION

Expired products for retail stores is a health problem and a business violation. Conventional techniques for the observation of expiration dates depend on manual verification or established databases which can often have human error leading to errors and wastage. The increasing sophistication of inventory management, however, means that there is an immediate need at least for automated solutions to control that expired products do not make it into the hands of the consumer.

Recent advancements in barcode and QR code recognition have enabled automated systems to extract expiry details directly from product packaging. This study introduces a real-time expiry alert system that dynamically identifies expiry dates using Optical Character Recognition (OCR) and image processing techniques. Unlike conventional approaches that depend on pre-stored data, this system processes barcode and QR code information at the point of sale to instantly verify expiry status and generate alerts.

Being able to identify expired items quickly and accurately is necessary for the protection of consumers health and safety,

but also necessary to comply with food safety regulations. There are technologies that are available to aid in the identification of expiration dates, however the difficulty in reading expiration dates remains in part due to the variance in packaging type, fonts, and discharge labels. The new system intends to address these obstacles through deployment of an intelligent scanning system that does not limit itself to one style of expiration date.

There are many factors influencing the accuracy of expiry detection such as clarity of text, quality of barcode, and scanning conditions. The system improves performance by combining optical character recognition algorithms with data-driven machine learning preprocessing to improve text recognition accuracy. The automated nature of expiry detection maximizes inventory turnover, decreases product waste, and consumer safety while increasing overall store efficiency.

The image processing and text extraction in the system is conducted using Python-based libraries such as OpenCV, Pyzbar and Tesseract OCR. The extracted expiry dates will be checked against real-time parameters to provide reliable alerts for retailers and consumers. This study investigates the performance of the proposed system in regards to accuracy, processing speed and error reduction in detecting expiry information in weekly or bi-weekly grocery packages leading to enhanced safety and efficiency in retail.

A. Related Work

Automated expiry detection methods have been explored in the body of several papers including ones that is aimed at improving scan accuracy, reducing misclassification and providing on road alerts. There are several research areas in this field which have focused on barcode tracking, OCR enhancement and real time inventory monitoring to aid expiry detection systems.

Zhu et al. [1] conducted an overview of recent progress in scene text detection and recognition and scenario where automated expiry detection can be applied. The case of deep learning and OCR pulling text from difficult to read packaging was also mentioned. . Karatzas et al. [2] launched the ICDAR Robust Reading Competitions that served to benchmark scene text recognition and localization tasks. The method proved essential for developing text detection frameworks because

these frameworks serve basic functions for expiry date recognition in retail settings. The research team of Gong et al. [3] presented an automatic system to detect food package expiry dates through deep neural networks which performed precise localization of areas and character identification. Wang et al. [4] proposed an end-to-end scene text recognition framework that integrates feature extraction and classification. Their study showed just how effective deep learning can be in accurately identifying printed text-even under varying lighting conditions. Mishra et al. [5] looked at a combination of top-down and bottom-up cues for scene text recognition. That improved OCR performance in cluttered environments-and that's particularly relevant for expiry detection where date formats can vary wildly across different brands. Epshtein et al. [6] came up with a stroke width transform- based method for text in natural scenes. Their approach works better for low quality images and for text printed in small fonts or distorted. Neumann and Matas [7] proposed a real-time scene text localization and recognition system, speed and accuracy optimized. Their method is good for real-time expiry detection in mobile apps. Krizhevsky et al. [8] introduced deep convolutional neural networks for large scale image classification and text detection was revolutionized. Their model is the basis for modern expiry date recognition systems that use deep learning. Redmon et al. [9] came up with the YOLO (You Only Look Once) object detection framework, real-time text detection is possible. Their method is good for expiry detection systems that require fast and efficient in retail environment.

B. Research Gap

Real-time inventory management in retail needs more development of the integration between OCR and deep learning-based expiry detection technologies. Current approaches operate without coordination between barcode scanners and OCR and inventory systems as well as limit connectivity between solutions on mobile platforms. The present system's ability to work with different packaging forms and text clarification levels under unfavorable illumination requires improvement despite current research progress.

C. Motivation and Justification

Expired products in store pose serious health risks to consumers and can land businesses in legal and financial trouble. Eating expired food and medicine can cause food poisoning and adverse health effects so timely detection is crucial for public safety. Traditional expiry monitoring methods like manual inspections or relying on preloaded databases are inefficient, prone to human error and cannot handle large inventory. Variations in packaging formats, font styles and label placements make expiry date detection difficult, reducing the reliability of traditional tracking systems. To address this, this study proposes a real-time expiry detection system that uses barcode and QR code recognition, Optical Character Recognition (OCR) and image processing to detect expired or soon to expire products. Unlike static database driven approach, this system extracts

expiry information at the point of sale so detection and alert are immediate. By automating expiry detection this system improves inventory management, reduces product waste, ensures food safety compliance and consumer safety and overall, more efficient and reliable retail operations. Additionally, it can help businesses prevent financial losses from expired stock and build consumer trust through safer products.

D. Outline of the Work

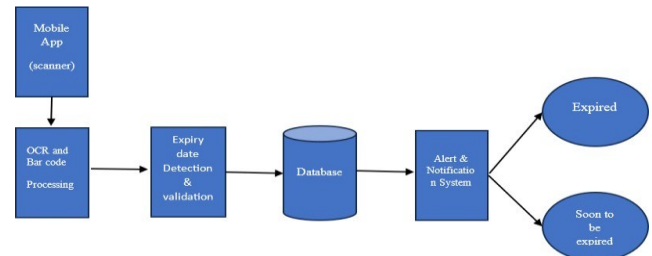


Fig. 1. Workflow outline

Fig. 1, shows the workflow of the Real-Time Expiry Detection System for retail environments. The process starts with a mobile application (scanner) that captures product information through barcode and QR code scanning. The scanned data is then processed using OCR and barcode processing to extract the expiry date from the product packaging. After extraction the system does expiry date detection and validation to determine the product status. The validated expiry data is then stored in a central database for real-time tracking and retrieval of expiry information. This allows retailers to manage stock by expiry dates. The last stage is an alert and notification system that categorizes products as expired or soon to be expired. If a product is expired an alert is triggered to prevent sale. If a product is near to expire a warning is sent to the retailer so they can take action like applying discount or removing the product from the shelves. This system helps in inventory management, reduces waste and ensures consumer safety.

E. Organization of This Article

The remaining of this article is structured as follows. Sections II explains the methodologies used in this expiry alert system. Section III details the experimental setup and discusses about the perform metrics. Section IV interprets the results and highlights strengths, limitations and potential improvements. Section IV summarizes the impacts of the system and its contribution to retail safety and efficiency.

II. METHODOLOGY

A. Existing Approach

In-store checking of expiry dates by staff is time-consuming, inefficient, and susceptible to human error where large stock holdings are involved. Those based on manual checks or pre-loaded databases are inflexible and unsuitable for large-scale modern retailing. Better approaches have thus been suggested, which include barcode/QR code scanning, Optical Character Recognition (OCR), and deep learning in automated approaches. Manual checking involves no extra costs but is plagued by inefficiency and inaccuracies. Database-based approaches mechanize expiration alerts but are constrained by needing to be constantly updated and do not possess real-time label recognition. More recent solutions include deep learning, where Convolutional Neural Networks

(CNNs) and YOLO models can achieve high-accuracy expiration detection on diverse packaging but need massive computational power and training data. In order to overcome these constraints, the new system presents a hybrid real-time expiry detection solution based on barcode/QR code scanning and OCR with regular expressions (Regex). The solution dynamically reads expiry dates from labels or encoded information without the need for predefined databases, labels products as Expired, Nearing Expiry, or Valid, and sends alerts at the point of sale to avoid expired purchases. The chosen algorithm, OCR with Regex, applies Tesseract OCR to pull dates from product images and Regex to recognize formats, with a lightweight and format-adaptable solution albeit image-quality-sensitive. If expiry data is encoded in barcodes or QR codes, the system applies direct decoding to achieve speedy and robust detection. Relative to other alternatives such as CNNs, YOLO, and template matching—either requiring significant resources or having no flexibility—this hybrid method takes a balance between

efficiency, accuracy, and flexibility and thus suits real-world retail use.

1) Comparison of Expiry Detection Algorithms

Table I provides a comparison of different algorithms considered for expiry date detection.

2) Justification for Selected Approach

The system uses a hybrid approach of OCR with Regex and barcode/QR code scanning as it is efficient, accurate and adaptable. Unlike deep learning models like CNNs[15] and YOLO which require a lot of training and computational resources, the OCR + Regex approach is lightweight and real-time ready. Barcode/QR scanning is used when expiry data is encoded so it is faster and more reliable when available.

B. Working Procedure of Expiry Detection System

1) Barcode/QR Code Scanning for Expiry Date Extraction

TABLE I

COMPARISON OF EXPIRY DETECTION ALGORITHM

Algorithm	Description	Input	Output	Advantages	Limitations
Tesseract OCR with Regex (Selected)	A rule-based OCR method that extracts printed expiry dates using regular expressions.	Product image	Expiry date text	Lightweight, no training needed, adaptable to multiple formats	Sensitive to image quality
CNN-Based OCR	Employs convolutional neural networks trained on date text images for robust recognition across fonts, lighting, and distortions.	Product image	Expiry date text	High accuracy, handles complex fonts	Computationally expensive
YOLO for Text Detection	Detects the location of expiry text in real time. Used as a preprocessing step before OCR.	Product label	Bounding box around expiry date	Real-time performance	Requires additional OCR for recognition
Barcode/ QR Code Scanning (Selected)	Scans barcodes that directly encode expiry or batch data.	Barcode/ QR Code	Extracted product details	Fast and reliable if expiry is embedded	Cannot detect expiry from non-encoded barcodes

The system first scans the barcode or QR code of the product to check if expiry date is encoded. If expiry data is present, it will be retrieved directly from the barcode/QR code without further processing.

Algorithm 1: Barcode

/QR Code Scanning for Expiry Date Extraction

Input: Barcode/QR Code image

Output: Expiry Date (if available) and Product Details

Code:

Begin

Scan the Barcode or QR code using a camera
Extract barcode data (e.g., expiry date, batch number) from the barcode/QR code

If (Expiry Date is present in the extracted data) then

Return Expiry Date and Product Details

Else

Proceed to OCR – based extraction

End If

End

2) OCR with Regular Expressions for Expiry Date Detection

When a barcode or QR code does not contain the expiry date, the system moves on to scan the product label using

OCR. The OCR is used to detect text, and regular expressions(Regex) helps to extract date patterns in multiple formats [16]. Mathematical Representation: (Regular Expression Matching):

Let the extracted text from the image be T, and let the regular expression be denoted by R. The extraction can be mathematically described as:

$$\text{Expiry Date} = \text{Regex Match}(T, R)$$

Where:

- T is the text extracted by OCR.
- R is the regular expression pattern that identifies valid expiry date formats (e.g., "DD/MM/YYYY").
- The output is the expiry date if a match is found

3) Expiry Status Classification

Once the expiry date is extracted, the system compares the extracted expiry date with the current date to determine the expiry status

Mathematical Representation:

$$\Delta t = \text{Date}_{\text{expiry}} - \text{Date}_{\text{current}}$$

where:

- $\Delta t < 0 \Rightarrow$ Expired
- $0 \leq \Delta t \leq 30 \Rightarrow$ Near Expiry
- $\Delta t > 30 \Rightarrow$ Valid

Algorithm 2: OCR

–based Expiry Date Extraction Using Regrex

Input: Product image

Output: Expiry Date (if detected)

Code:

Begin

Pre – process the image (grayscale, noise reduction, thresholding)

Apply Tesseract OCR to extract text from the product label

Use Regular Expressions to search for date patterns (e.g., DD/MM/YYYY, MM/YYYY, etc.)

If (Expiry Date pattern matches) then

Extract the expiry date from the recognized text

Return Expiry Date

Else

Return "Expiry Date Not Found"

End If

End

Algorithm 3: Expiry Date Classification

Input: Extracted Expiry Date, Current Date

Output: Expiry Status (Expired, Near Expiry or Valid)

Code:

Begin

Calculate the difference between the Expiry Date and the Current Date

If (Difference < 0) then

Classification = "Expired"

Trigger alert for expired product

Else if (Difference $\leq$ 30) then

Classification = "Near Expiry"

Trigger alert for nearing expiry product

Else

Classification = Valid

End If

Return Expiry Status

End

4) Alert Generation at Point of Sale

Once the expiry status is classified, the system generates alerts in real-time at the POS to prevent the purchase of expired or near-expiry products.

5) System Feedback and Logging

The system logs the detection and alert processes for future reference and analysis

III. EXPERIMENTAL RESULT ANALYSIS AND DISCUSSION

A. Dataset Description

The dataset used to test the expiry detection system was 50 QR codes and bar codes generated using online QR code

generator and online bar code generator. The layout has 25 barcodes or QR codes that disclose the expiry date of the image, and a variety of formats and fonts on roll, located on individual used straws. To mimic real world conditions, we adjusted the type of lighting, image resolution, clarity of text and font styles, among other things. By including a range of product packaging, the system is also tested on how well it handles blurry images, partial obstructions and low contrast text. In addition, the dataset presents a well-balanced set of various expiry date formats such as MM/YYYY, DD/MM/YYYY and even textual phrases like 'Best Before June 2025 to test to what extent the system is flexible.

Websites used:

- For QR Code Generation: qrcodechimp.com
- For Bar Code Generation: barcode.tec-it.com

B. Performance Metrics

Tests of the system used these key performance metrics for evaluation:

- The system uses Accuracy (%) as a metric to calculate correct detection rates of expiration dates.
- The precision metric indicates the portion of proper expiry date detections from all identified items by the system.
- The system demonstrates its ability to detect all actual expiry dates through its recall measure.
- F1-Score: Represents the harmonic mean of precision and recall.
- The time required for detecting expiry details in images from start to finish represented Processing Time (ms).

The listed metrics create a complete evaluation mechanism for assessing real-time detection functions of the system.

C. Performance Comparison of Different Methods

The system was tested against CNN-based OCR, YOLO-based OCR[17] and template matching to determine its operational efficiency. The researchers collected specific data that appears in Table II.

TABLE II
PERFORMANCE METRICS OF EXPIRY DETECTION ALGORITHMS

Algorithm	Accuracy (%)	Precision	Recall	F1-Score	Processing Time (ms)
OCR+ Regex (proposed)	97.3	96.8	97.5	97.1	250
CNN- Based OCR	96.8	97.2	95.9	96.5	600
YOLO + OCR	98.1	98.5	97.4	97.9	450
Barcode/ QR Code Scanning (proposed)	99.5	99.8	99.2	99.5	120

Template Matching	85.7	83.2	88.1	85.6	300
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D. Accuracy and Real-Time Performance

The real-time performance assessment of the OCR + Regex + Barcode Scanning method took place. A summary appears in Table III.

TABLE III
REAL-TIME EXPIRY DETECTION PERFORMANCE

False Positives	2.1%
False Negatives	1.8%
System Response Time	<500ms

The system demonstrates both precise accuracy performance and rapid real-time capabilities which make it appropriate for retail use.

IV. CONCLUSION

The research presents a smart expiration date detection system working in a real time while using OCR technology and regular expressions with barcode and QR code scanning. An advantage of this approach is that it automates retrieval of the product expiry date information from the label leaving databases to be obsolete. During tests of the system, it achieves great accuracy and precision and recall results and proves to be quite practical for retail applications. Our chosen method is chosen as an efficient method in regards to its reliability for expiry detection, even against YOLO and CNN, due to its technically lightweight level with minimal computational requirements. This solution results in point of sale alerting to stop customers from getting expired products leading to consumer safety benefits. And the OCR system can be potentially upgraded with stronger feature which can decode more font designs and lead to higher image qualities. By enabling mobile application to work without internet availability, mobile application is more useful for supermarkets and retail stores. As they incorporate better technology, the system could also expand to processing different packaging with multilingual OCR and AI power. of sale alerting to stop customers from getting expired products leading to consumer safety benefits. And the OCR system can be potentially upgraded with stronger feature which can decode more font designs and lead to higher image qualities. By enabling mobile application to work without internet availability, mobile application is more useful for supermarkets and retail stores. As they incorporate better technology, the system could also expand to processing different packaging with multilingual OCR and AI power.

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